Does dynamic trust better predict escalation?

REFERENCE

MatrAIx: Simulating the World with 8.3 Billion Persona Agents

Li et al. (2026), arXiv:2608.04205

Large persona simulations for evaluating AI systems. We test one piece those personas usually freeze: trust while the conversation is still going.

THE PROBLEM

- A static agent keeps one trust score, so its tone does not change when the partner is dismissive.
- People lose patience faster than they regain it.
- If both agents stay polite, a blind judge can call the thread civil even when the real conversation later had a personal attack.

DATASET: CGA-WIKI

- 4,188 Wikipedia talk-page conversations (Zhang et al., ACL 2018).
- Two speakers. Agents see only the first 3 turns, cut off before any attack.
- Label: the real thread later had a personal attack, or it did not.
- Each attack thread is paired with a civil one from the same page.
- Later human turns and the label stay hidden from the agents.

OUR COMPARISON

- Same opening, personas, model, prompt, and length in both arms.
- Static: trust stays at the value set by rating the opening.
- Dynamic: the listener updates trust after every new message, and the next reply follows that mood.
- A blind judge scores the dialogue only. We then compare those scores with the real label.

Pipeline and progress

HOW A CONVERSATION RUNS

1 Real opening
First 3 comments. Nothing after that.

↓

2 Starting trust
Score those 3 comments. Both arms start here.

↓

3 Static
Trust stays fixed

3 Dynamic
Trust changes each turn

↓

4 Blind judge
Reads the dialogue only.

↓

5 Compare with the real label
Did this thread actually have an attack?

TRUST RULE

New trust = old trust + step × (rating − 0.5)

A rude reply lowers trust more than a kind reply raises it.
A forgives faster. B stays annoyed longer.

PILOT · 5 CONVERSATIONS

Real label	Static	Dynamic
Attack	0.72	0.75
Civil	0.00	0.00
Civil	0.00	0.00
Attack	0.00	0.00
Civil	0.35	0.20

Both arms: **AUC 0.67**

0.50 would be chance. Dynamic is not better than static yet. Five conversations only.

Next. 20 full pairs, then the test set.

One conversation, two arms

ONCE, BEFORE EITHER ARM

Both start at 0.75
Same prior for A and B.

→

Score the 3 real comments
Each person scores what the other wrote.

→

Copy those two scores
Static and dynamic start identical.

EVERY NEW MESSAGE

1. Speaker writes
Tone comes from their trust right now.

→

2. Listener scores it
0 rude · 0.5 neutral · 1 kind.

→

3. Only the listener moves
The speaker's trust stays put.

Then they swap. The listener speaks next.

Static
Same writers. Trust never moves again.

Dynamic
This loop runs after every reply.

Why they don't move together. A sharp opening can leave B lower than A. A kind reply later lifts only the listener, and only a little, because trust falls faster than it rises.